IP tarmoqni subnetlarga bo‘lish

IP Subnetting

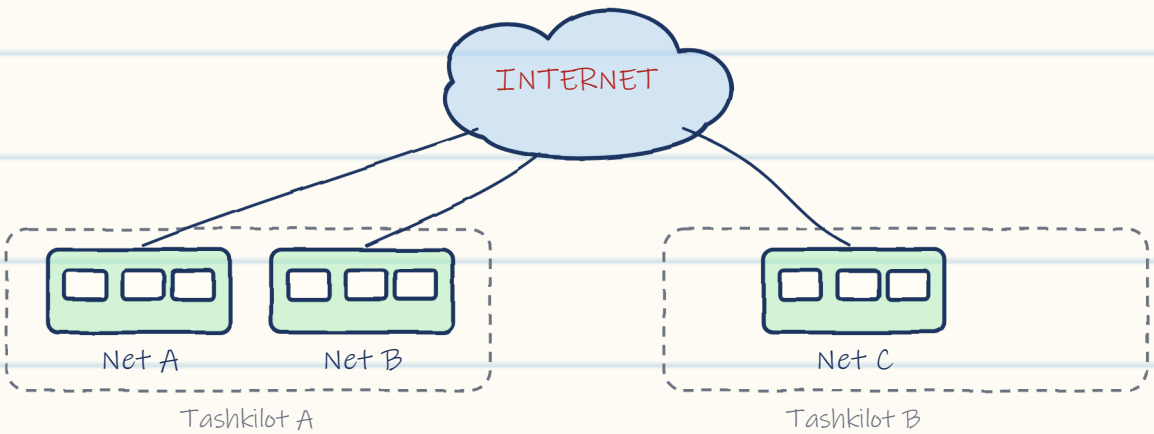
Asosiy atamalar

- Subnet — katta tarmoqni bo‘lib olingan kichik tarmoq
- Subnet maska — manzilning qaysi qismi tarmoq, qaysi qismi host ekanini ko‘rsatadi
- Prefiks (/24) — maskaning qisqa yozuvi — nechta bit tarmoqqa tegishli
- Qarzga olingan bit — host qismidan olinib, subnet raqamiga beriladigan bitlar
- VLSM — bitta tarmoqda turli uzunlikdagi maskalarni ishlatish
- CIDR — bir nechta tarmoqni bitta yirik prefiksga jamlash

1. Nega subnet kerak?

Dastlab manzil ikki pog‘onali: Net ID + Host ID. Har fizik tarmoqqa — alohida sinf manzili.

- manzil isrofi — 300 host uchun B sinfi (65534) beriladi
- tarmoq yetmaydi — 100 kichik tarmoqqa 100 ta C sinfi kerak
- kengaymaydi — yangi tarmoq uchun IANA’dan yangi manzil kutiladi



har fizik tarmoq — alohida sinf manzili

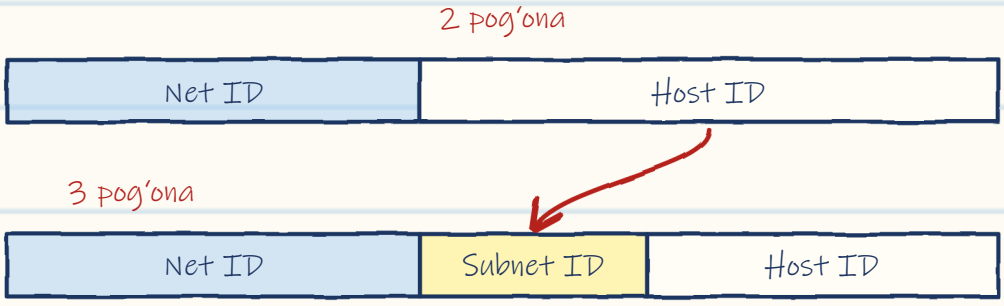
1-rasm. Sinf bo'yicha taqsimlash: tarmoqlar ko'p, manzil yetmaydi



yechim: host bitlaridan bir qismini subnetga berish

2. Subnetlash usuli

RFC 950 va RFC 917 — Host ID'dan bit qarzga olinadi va Subnet ID bo'ladi.



subnet tashqariga ko'rinmaydi — marshrutizator uni ajratadi

2-rasm. Net ID + Subnet ID + Host ID

3. Subnet maska

- 32 bit: ketma-ket 1 — tarmoq, ketma-ket 0 — host
- standart maskalar: A — 255.0.0.0, B — 255.255.0.0, C — 255.255.255.0

IP	10000110.10010000.00000001.00000001	134.144.1.1
maska	11111111.11111111.11111111.00000000	255.255.255.0
AND		
subnet	10000110.10010000.00000001.00000000	134.144.1.0
broadcast	10000110.10010000.00000001.11111111	134.144.1.255

host bitlari hammasi 0 → subnet manzili

host bitlari hammasi 1 → broadcast manzili

3-rasm. IP AND maska = subnet manzili

4. Subnetdagi host manzillari

Host bitlari N bo'lsa, foydali manzil $2^N - 2$ ta.

192.168.3.192 · maska 255.255.255.224



maska: 11111111.11111111.11111111.11100000

tarmoq bitlari 27 → N = 32 − 27 = 5

host manzillari = $2^5 - 2 = 30$

subnet .192 · hostlar .193...222 · broadcast .223

4-rasm. /27 uchun hisob

2^n	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8	2^9	2^{10}
=	2	4	8	16	32	64	128	256	512	1024

5. Host soniga qarab bo'lish

Har subnetda Y manzil kerak: $2^N \geq Y+2 \geq 2^{(N-1)}$, maska uzunligi 32 − N.

168.195.0.0 · Y = 700 → N = 10



/22 = 255.255.252.0

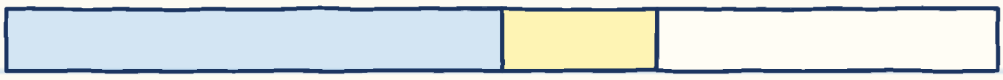
2^6 = 64 subnet: 168.195.0.0 · 4.0 · 8.0 · 12.0 ...

5-rasm. 700 host → /22

6. Subnet soniga qarab bo'lish

X ta subnet kerak: $2^M \geq X \geq 2^{(M-1)}$ — hostlar soni maksimal qoladi.

168.195.0.0 → 27 ta subnet, M = 5



16 + 5 = 21 → /21 = 255.255.248.0

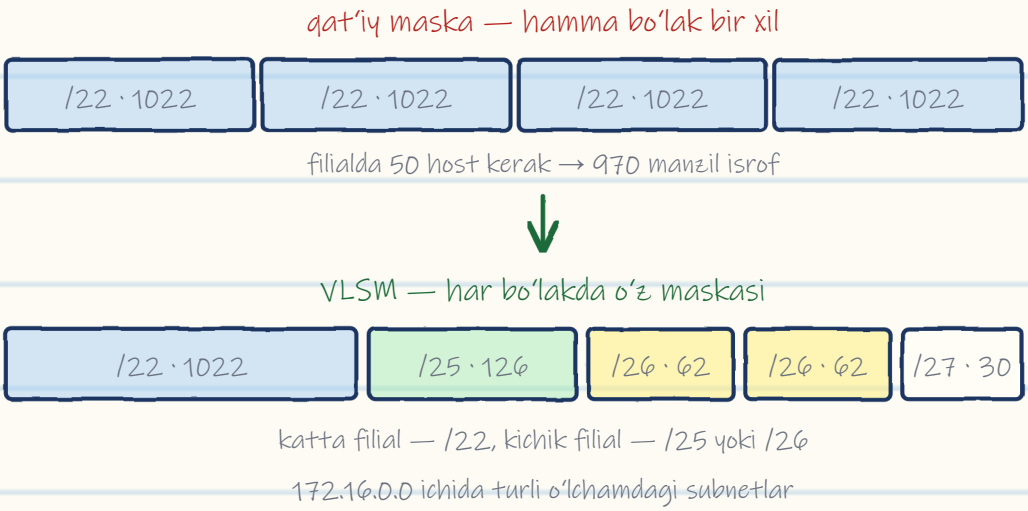
168.195.0.0 | 168.195.8.0 | 168.195.16.0 | 168.195.24.0 ...

jami 2^5 = 32 subnet — 27 tasi yetarli

6-rasm. 27 subnet → /21

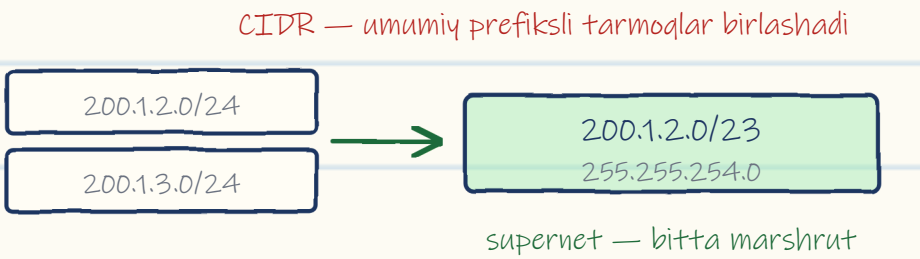
7. VLSM

- bitta maskali bo'lish — hamma subnet bir xil, manzil isrof bo'ladi
- RFC 1009 — turli uzunlikdagi maskalar, bo'laklar har xil kattalikda



8. CIDR

1992: B sinfi tugayapti, marshrut jadvali o'sdi. Yechim — sinfsiz prefiks (RFC 1517–1520).



slash yozuvi: 192.168.1.1/27



maskadagi birlar soni = prefiks uzunligi

yig'ish (aggregation) marshrut jadvalini qisqartiradi

7-rasm. Supernet va /prefiks yozuvi

Xulosa

- Subnet = Host ID'dan qarzga olingan bitlar → 3 pog'ona
- Maska: 1 lar — tarmoq, 0 lar — host; IP AND maska = subnet
- Host soni = 2^N − 2; subnet soni = 2^M
- Y host kerak → 2^N ≥ Y+2; X subnet kerak → 2^M ≥ X
- VLSM — turli maska; CIDR — sinfsiz prefiks va supernet

Tekshirish

1. /27 da nechta host? — 2^5 − 2 = 30
2. 134.144.1.1/24 broadcast'i? — 134.144.1.255
3. 700 host uchun maska? — /22 = 255.255.252.0
4. VLSM qaysi RFC? — RFC 1009